# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The probability distribution of a random variable $X$ is $PX = \begin{matrix} \frac{X}{12} & : & X = 1,2,3,4,5,6 \\ 0 & : & \text{otherwise} \end{matrix}$. Then, the conditional probability $P\frac{\frac{3}{2} < X < \frac{7}{2}}{X > 2}$ is

**(A)** $\frac{5}{6}$

**(B)** $\frac{5}{18}$

**(C)** $\frac{1}{6}$

**(D)** $\frac{7}{12}$

**Q.2**

A certain type of missile hits the target with probability $\mathrm{p} = 0.3$. What is the least number of missiles should be fired so that there is at least an $80\%$ probability that the target is hit?

**(A)** 5

**(B)** 6

**(C)** 7

**(D)** None of the above

**Q.3**

Seven white balls and three black balls are randomly placed in a row. The probability that no two black balls are placed adjacently equals

**(A)** $\frac{1}{2}$

**(B)** $\frac{7}{15}$

**(C)** $\frac{2}{15}$

**(D)** $\frac{1}{3}$

**Q.4**

A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is

**(A)** $8/729$

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**(B)** $8/243$

**(C)** $1/729$

**(D)** $8/9$.

**Q.5**

In a sequence of independent trials, the probability of success on each trial is $1/4$. The probability that the second success occurs on the fourth or later trial, if the trials continue up to the second success only, is

**(A)** $\frac{5}{32}$

**(B)** $\frac{27}{32}$

**(C)** $\frac{23}{32}$

**(D)** $\frac{9}{32}$

**Q.6**

6 coins are tossed together 64 times. If throwing a head is considered as a success then the expected frequency of at least 3 successes is

**(A)** 64

**(B)** 21

**(C)** 32

**(D)** 42

**Q.7**

The probability of a man hitting a target is $1/4$. The number of times he must shoot so that the probability he hits the target, at least once is more than $0.9$, is $[$ use $\log 4 = 0.602$ and $\log 3 = 0.477]$

**(A)** 7

**(B)** 8

**(C)** 6

**(D)** 5

**Q.8**

In a binomial distribution $B\left(n, p = \frac{1}{4}\right)$, if the probability of at least one success is greater than or equal to $\frac{9}{10}$, then $n$ is greater than:

**(A)** $\frac{1}{\log_{10} 4+\log_{10} 3}$

**(B)** $\frac{9}{\log_{10} 4-\log_{10} 3}$

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**(C)** $\frac{4}{\log_{10} 4-\log_{10} 3}$

**(D)** $\frac{1}{\log_{10} 4-\log_{10} 3}$

**Q.9**

A consignment of 15 washing machines contains 4 defective ones. The washing machines are selected one by one and examined. The examined ones are not restored. If the probability of $9^{\text{th}}$ one examined is the last defective is $\frac{a}{b}$ (where $a$ and $b$ are co-prime), then the value of $\frac{b-3}{a}$ is

**(A)** 53

**(B)** 24

**(C)** 78

**(D)** 89

**Q.10**

If $X$ follows a binomial distribution with parameters $\mathrm{n}=8$ and $\mathrm{p}=\frac{1}{2}$, then $\mathrm{P}(|\mathrm{X}-4| \leq 2)$ is

**(A)** $\frac{119}{128}$

**(B)** $\frac{119}{228}$

**(C)** $\frac{19}{128}$

**(D)** $\frac{18}{128}$

**Q.11**

There are two bags, bag 1 contains 3 white and 2 red balls and bag 2 contains 4 white and 5 red balls. A ball is drawn randomly from bag 1 and put in bag 2. Now 2 balls are drawn from bag 2 and found to be red. Then the probability that white ball was drawn from bag 1 to bag 2 is

**(A)** $< \frac{1}{2}$

**(B)** $< \frac{1}{4}$

**(C)** $> \frac{3}{4}$

**(D)** $> \frac{1}{4}$

**Q.12**

If mean and variance of a Binomial variate $\mathrm{X}$ are 2 and 1 respectively, then the probability that $\mathrm{X}$ takes a value greater than 1 is

**(A)** $\frac{2}{3}$

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**(B)** $\frac{4}{5}$

**(C)** $\frac{7}{8}$

**(D)** $\frac{11}{16}$

**Q.13**

If $A$ and $B$ are two independent events such that $\mathrm{P}\left(\bar{\mathrm{A}} \cap \mathrm{B}\right) = \frac{2}{15}$ and $\mathrm{P}\left(\mathrm{A} \cap \bar{\mathrm{B}}\right) = \frac{1}{6}$ and $p_1, \quad p_2$ are possible values of $\mathrm{P}(\mathrm{B})$, then evaluate $\frac{1}{p_1} + \frac{1}{p_2}$.

**(A)** 9.25

**(B)** 5.50

**(C)** 7.25

**(D)** 8.80

**Q.14**

If the random variable $X$ takes the values $x_1, x_2\ x_3, \ldots, x_{10}$ with probabilities $P\left(X = x_i\right) = ki$, then the value of $\mathrm{k}$ is equal to

**(A)** $\frac{1}{10}$

**(B)** $\frac{1}{15}$

**(C)** $\frac{1}{55}$

**(D)** 10

**Q.15**

The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is

**(A)** $\frac{28}{256}$

**(B)** $\frac{219}{256}$

**(C)** $\frac{128}{256}$

**(D)** $\frac{37}{256}$

**Q.16**

A random variable $X$ has the probability distribution:

| X : | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| P(X) : | 0.15 | 0.23 | 0.12 | 0.10 | 0.20 | 0.08 | 0.07 | 0.05 |

For the events $E = \{X$ is a prime number $\}$ and $F = \{X < 4\}$, the probability $P(E \cup F)$ is

**(A)** 0.87

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**(B)** $0.50$

**(C)** $0.77$

**(D)** none of these

**Q.17**

The probability that the screws are defective in three boxes of screws $A$, $B$ and $C$ are $\frac{1}{5}$, $\frac{1}{6}$ and $\frac{1}{7}$ respectively. A box is selected at random and a screw drawn from it randomly is found to be defective. The probability that the defective screw is drawn from the box $A$ is

**(A)** $\frac{16}{29}$

**(B)** $\frac{1}{15}$

**(C)** $\frac{27}{59}$

**(D)** $\frac{42}{107}$

**Q.18**

A random variable has the following probability distribution

| x : | o | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| p(x) | o | 2p | 2p | 3p | $p^2$ | $2p^2$ | $7p^2$ | $2p$ |

The value of p is

**(A)** $\frac{1}{10}$

**(B)** $-1$

**(C)** $\frac{3}{10}$

**(D)** none

**Q.19**

The mode of the binomial distribution for which mean and standard deviation are 10 and $\sqrt{5}$ respectively is

**(A)** 7

**(B)** 8

**(C)** 9

**(D)** 10

**Q.20**

A binomial variate $X$ has mean $= 6$ and variance $= 2$ the probability that $5 \leq X \leq 7$ is

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**(A)** $\frac{4}{2}$

**(B)** $\frac{1622}{6661}$

**(C)** $\frac{4672}{6561}$

**(D)** none

## Answers & Solutions

**Q.1 Answer:**

$\frac{1}{6}$

**Solution:**

**Let,** $A = \frac{3}{2} < X < \frac{7}{2}, \quad B = X > 2$

$A \cap B = 2 < X < \frac{7}{2}$

$P\frac{A}{B} = \frac{PA \cap B}{PB} = \frac{P3}{P3 + P4 + P5 + P6}$

$= \frac{\frac{3}{12}}{\frac{3}{12} + \frac{4}{12} + \frac{5}{12} + \frac{6}{12}} = \frac{1}{6}$

**Q.2 Answer:**

5

**Solution:**

**Probability of hitting the target** $= 0.3$ **If 'n' is the no. of times that the Missile is fired.**

$\therefore$ **Probability of hitting at least once** $= 1 - [1 - 0.3]^{n} = 0.8$

$0.7^{n} = 0.2$

$n \log 0.7 = \log 0.2$

$\Rightarrow \quad n = 4.512$

**for** $n = 4; p < 0.8$

**take** $n = 5$

$n = 5$

**Hence 5 missiles should be fired so that there is at least** $80\%$ **prob. that the target is hit.**

**Q.3 Answer:**

$\frac{7}{15}$

**Solution:**

**The number of ways of placing 3 black balls without any restriction is** ${}^{10}C_3$.

**Since, we have total 10 places of putting 10 balls in a row.**

**Now, the number of ways in which no two black balls put together is equal to the number of ways of choosing 3 places** $m$ **arked '-' out of eight places**

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$W - W - W - W - W - W - W - W -$

**This can be done in ${}^{8}C_{3}$ ways**

$\therefore \quad$ Required probability $= \frac{{}^{8}C_{3}}{{}^{10}C_{3}} = \frac{8\times7\times6}{10\times9\times8} = \frac{7}{15}$

**Q.4 Answer:**

$8/243$

**Solution:**

**A pair of fair dice is thrown, the sample space** $S = (1,1),(1,2)(1,3)\ldots = 36$

**Possibility of getting 9 are** $(5,4),(4,5),(6,3),(3,6)$

**- Possibility of getting score 9 in a single throw** $= \frac{4}{36} = \frac{1}{9}$

**Probability of getting score 9 exactly twice** $= {}^{3}\mathrm{C}_{2} \times \left(\frac{1}{9}\right)^{2} \cdot \left(1 - \frac{1}{9}\right) = \frac{3!}{2!} \times \frac{1}{9} \times \frac{1}{9} \times \frac{8}{9} = \frac{8}{243}$

**Q.5 Answer:**

$\frac{27}{32}$

**Solution:**

**Let the second success occur at the** $\mathrm{nth}$ **trial. This means that there was exactly one success in the first** $\mathrm{n} - 1$ **trials, so that the probability of getting the second success at the nth trial is**

$p_n = \left({}^{n-1}C_1 pq^{n-1-1}\right) p = (n-1)p^2q^{n-2}$

**Therefore the probability of the required event is** $\mathrm{p}_4 + \mathrm{p}_5 + \mathrm{p}_6 + \ldots = 3\mathrm{p}^2\mathrm{q}^2 + 4\mathrm{p}^2\mathrm{q}^3 + 5\mathrm{p}^2\mathrm{q}^4 + 6\mathrm{p}^2\mathrm{q}^5 + \ldots$

$= p^2q^2\left(3 + 4q + 5q^2 + 6q^3 + \ldots\right)$

$= \mathrm{p}^2\mathrm{q}^2\left[3\left(1 + \mathrm{q} + \mathrm{q}^2 + \mathrm{q}^3 + \ldots\right) + \mathrm{q}\left(1 + 2\mathrm{q} + 3\mathrm{q}^2 + \ldots\right)\right]$

$= p^2q^2\left[3(1-q)^{-1} + q(1-q)^{-2}\right] = p^2q^2\left(\frac{3}{p} + \frac{q}{p^2}\right)$

$= q^2(3p + q) = q^2(2p + p + q) = q^2(2p + 1)$

$= \left(\frac{3}{4}\right)^2\left[2\left(\frac{1}{4}\right) + 1\right] = \frac{9}{16}\left(\frac{1}{2} + 1\right) = \frac{27}{32}$

**Q.6 Answer:**

42

**Solution:**

**Here** $\mathrm{p} = \frac{1}{2}.\mathrm{q} = 1 - \mathrm{p} = 1 - \frac{1}{2} = \frac{1}{2}$

$\mathrm{n} = 6,\ \mathrm{N} = 64.$

**Then**

$p(r) = {}^{n}C_r p^r q^{n-r} = {}^{6}C_r\left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{6-r} = {}^{6}C_r\left(\frac{1}{2}\right)^6$

$\therefore \mathrm{f(r)} = \mathrm{Np(r)} = 64.{}^{6}\mathrm{C}_{\mathrm{r}} \cdot \frac{1}{64} = {}^{6}\mathrm{C}_{\mathrm{r}}$

**Now** $\sum_3^6 \mathrm{p(r)} = \mathrm{p(3)} + \mathrm{p(4)} + \mathrm{p(5)} + \mathrm{p(6)}$

$= \left({}^{6}\mathrm{C}_3 + {}^{6}\mathrm{C}_4 + {}^{6}\mathrm{C}_5 + {}^{6}\mathrm{C}_6\right)\frac{1}{2^6}$

$= \left(2^6 - {}^{6}\mathrm{C}_0 - {}^{6}\mathrm{C}_1 - {}^{6}\mathrm{C}_2\right)\frac{1}{2^6}$

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$= (64 - 1 - 6 - 15)\frac{1}{2^6} = \frac{42}{64} = \frac{21}{32}$
$\therefore f(r)_{r\geq 3} = N\sum_3^6 p(r) = 64 \cdot \frac{21}{32} = 42$

**Q.7 Answer:**

8

**Solution:**

**Let n denote the required number of shots and $X$ the number of shots that hit the target. Then**
$X \sim B(n, p)$, **with** $p = 1/4$. **Now,** $P(X \geq 1) \geq 0.9 \Rightarrow 1 - P(X = 0) \geq 0.9$
$\Rightarrow 1 - {}^nC_0 \left(\frac{3}{4}\right)^n \geq 0.9 \Rightarrow \left(\frac{3}{4}\right)^n \leq \frac{1}{10}$
$\Rightarrow \left(\frac{4}{3}\right)^n \geq 10 \Rightarrow n(\log 4 - \log 3) \geq 1$
$\Rightarrow n(0.602 - 0.477) \geq 1 \Rightarrow n \geq \frac{1}{0.125} = 8$
**Therefore the least number of trials required is 8**

**Q.8 Answer:**

$\frac{1}{\log_{10} 4 - \log_{10} 3}$

**Solution:**

**We have** $P(x \geq 1) \geq \frac{9}{10}$
$\Rightarrow 1 - P(x = 0) \geq \frac{9}{10}$
$\Rightarrow 1 - {}^nC_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^n \geq \frac{9}{10}$
$\Rightarrow 1 - \frac{9}{10} \geq \left(\frac{3}{4}\right)^n \Rightarrow \left(\frac{3}{4}\right)^n \leq \left(\frac{1}{10}\right)$
**Taking log to the base** $3/4$, **on both sides, we get** $n \log_{3/4}\left(\frac{3}{4}\right) \geq \log_{3/4}\left(\frac{1}{10}\right)$
$\Rightarrow \quad n \geq -\log_{3/4} 10 = \frac{-\log_{10} 10}{\log_{10}\left(\frac{3}{4}\right)}$
$\Rightarrow n \geq \frac{1}{\log_{10} 4 - \log_{10} 3}$

**Q.9 Answer:**

24

**Solution:**

$A$: **Finding three defective ones in the examination of first** $8$ **washing machines.**
$B$: $9^{\text{th}}$ **washing machine is defective.**
**Required probability** $= P(A \cap B)$
$P(A \cap B) = P(A) \quad P(B \mid A) \quad \ldots (i)$
$P(A) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8}$
$P(B \mid A) = \frac{1}{7}$ **...[There is only one defective machine left in remaining last seven machines to be examined.]**
**Substituting** $P(A)$ **and** $P(B \mid A)$ **in** $(i)$, **we get**
**Required probability** $= \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7}$
$= \frac{4 \times (11 \times 10 \times 9 \times 8 \times 7) \times (1 \times 2 \times \ldots \times 7 \times 8)}{(15 \times 14 \times \ldots \times 9 \times 8) \times (1 \times 2 \times 3 \times 4 \times 5)} \times \frac{1}{7}$

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$= \frac{8}{195}$

$\Rightarrow a = 8, \quad b = 195$

$\Rightarrow \frac{b-3}{a} = \frac{192}{8} = 24$

**Q.10 Answer:**

$\frac{119}{128}$

**Solution:**

**We have,** $P(|X-4| \leq 2) = P(-2 \leq X-4 \leq 2) = P(2 \leq X \leq 6)$

$= 1 - [P(X=0) + P(X=1) + P(X=7) + P(X=8)]$

$= 1 - \left[{}^8C_0 \left(\frac{1}{2}\right)^8 + {}^8C_1 \left(\frac{1}{2}\right)^8 + {}^8C_7 \left(\frac{1}{2}\right)^8 + {}^8C_8 \left(\frac{1}{2}\right)^8\right]$

$= 1 - \left(\frac{1}{2}\right)^8 (1+8+8+1) = 1 - \frac{18}{2^8} = \frac{119}{128}$

**Q.11 Answer:**

$> \frac{1}{4}$

**Solution:**

$A_1$: **white ball is drawn from bag**

$A_2$: **Red ball is drawn from bag** 1

$B$: 2 **balls are drawn from bag** 2 **and found to be red.**

$P(B) = PA_1 \cdot PB / A_1 + PA_2 PB / A_2$

$\Rightarrow PA_1 / B = \frac{\frac{3}{5} \cdot \frac{{}^5C_2}{{}^{10}C_2}}{\frac{3}{5} \cdot \frac{{}^5C_2}{{}^{10}C_2} + \frac{2}{5} \cdot \frac{{}^6C_2}{{}^{10}C_2}} = \frac{1}{2}$

**Q.12 Answer:**

$\frac{11}{16}$

**Solution:**

**We have; np =** $2 =$ **mean** $npq = 1 =$ variance

$\Rightarrow p = \frac{1}{2}; q = \frac{1}{2} \& n = 4$

**Required probability** $= P(x > 1)$

$= 1 - P(x \leq 1)$

$= 1 - [P(x=0) + P(x=1)]$

$= 1 - \left[{}^4C_0 q^4 + {}^4C_1 q^3 p^1\right]$

$= 1 - \frac{5}{16} = \frac{11}{16}$

**Q.13 Answer:**

7.25

**Solution:**

**Let** $P(A) = x, \quad P(B) = y$

$P(\bar{A} \cap B) = P(\bar{A}) P(B)$

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$\Rightarrow \frac{2}{15} = \left(1 - x\right)y$

$\Leftrightarrow \frac{2}{15} = y - xy$ ... i

$P\left(A \cap \bar{B}\right) = P\left(A\right)P\left(\bar{B}\right)$

$\Rightarrow \frac{1}{6} = x\left(1 - y\right)$

$\Leftrightarrow \frac{1}{6} = x - xy$ ... ii

(ii) - (i) **gives**

$x - y = \frac{1}{6} - \frac{2}{15} = \frac{1}{30}$

$\Rightarrow x = y + \frac{1}{30}$

**Substituting** $x$ **in** (i), **we get**

$y - y\frac{1}{30} + y = \frac{2}{15}$

$\Leftrightarrow 30y^2 - 29y + 4 = 0$

$\Leftrightarrow \left(6y - 1\right)\left(5y - 4\right) = 0$

$\Leftrightarrow y = \frac{1}{6}$ **or** $\frac{4}{5}$

$\Rightarrow \frac{1}{p_1} + \frac{1}{p_2} = 6 + \frac{5}{4} = 7.25$

**Q.14 Answer:**

$\frac{1}{55}$

**Solution:**

$\sum_{i=1}^{10} p\left(X = x_i\right) = 1$

$\Rightarrow \mathrm{k}(1 + 2 + \ldots + 10) = 1$

$\Rightarrow \mathrm{k}\frac{10 \times 11}{2} = 1 \Rightarrow \mathrm{k} = \frac{1}{55}$

**Q.15 Answer:**

$\frac{28}{256}$

**Solution:**

**Mean** $= np = 4$ **and variance** $= npq = 2$

$\therefore p = q = \frac{1}{2}$ **and** $n = 8$

$\therefore P(2 \text{ success}) = {}^8C_2 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^2 = \frac{28}{2^8} = \frac{28}{256}$

**Q.16 Answer:**

$0.77$

**Solution:**

$P(E) = 0.23 + 0.12 + 0.20 + 0.07$

$= 0.62, \mathrm{P}(\mathrm{F}) = 0.5, \mathrm{P}(\mathrm{E} \cap \mathrm{F}) = 0.35$

$\mathbf{P}(\mathrm{E} \cup \mathrm{F}) = \mathbf{P}(\mathrm{E}) + \mathbf{P}(\mathrm{F}) - \mathrm{P}(\mathrm{E} \cap \mathrm{F}) = 0.77$

**Q.17 Answer:**

$\frac{42}{107}$

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**Solution:**

**Let $S =$ screws drawn is found to be defective**

$B_1 = A$ **box is selected;** $PB_1 = \frac{1}{3}; \quad P\frac{S}{B_1} = \frac{1}{5}$

$B_2 = B$ **box is selected;** $PB_2 = \frac{1}{3}; \quad P\frac{S}{B_2} = \frac{1}{6}$

$B_3 = C$ **box is selected;** $PB_3 = \frac{1}{3}; \quad P\frac{S}{B_3} = \frac{1}{7}$

$$P\frac{B_1}{S} = \frac{\frac{1}{3}\times\frac{1}{5}}{\frac{1}{3}\times\frac{1}{5}+\frac{1}{3}\times\frac{1}{6}+\frac{1}{3}\times\frac{1}{7}}$$

$$= \frac{42}{107}$$

**Q.18 Answer:**

$\frac{1}{10}$

**Solution:**

$$\sum p(x) = 1 \Rightarrow 9p + 10p^2 = 1$$

$$\Rightarrow 10\mathrm{p}^2 + 9\mathrm{p} - 1 = 0 \Rightarrow \mathrm{p} = -1 \text{ or } \frac{1}{10}$$

$$\mathrm{p} = \frac{1}{10}$$

**Q.19 Answer:**

10

**Solution:**

**In binomial distribution,**

**mean** $= np = 10$, **variance** $= npq = 5 \therefore p = q = \frac{1}{2}$

**Let** $x$ **be the mode, then** $np + p > x > np - q$

$$\therefore 10 + \tfrac{1}{2} > x > 10 - \tfrac{1}{2} \quad \Rightarrow \tfrac{21}{2} > x > \tfrac{19}{2}$$

$$\Rightarrow 9.5 < x < 10.5 \quad \Rightarrow x = 10.$$

**Q.20 Answer:**

$\frac{4672}{6561}$

**Solution:**

**Here mean** $= np = 6$; **variance** $= npq = 2$

$$\therefore \mathrm{q} = \tfrac{2}{9} = \tfrac{1}{3}; \quad \therefore \mathrm{p} = 1 - \mathrm{q} = \tfrac{2}{3}; \quad \therefore \mathrm{n} = 9$$

**Now, probability of** $5 \le x \le 7$ **is equal to**

$$p(5) + p(6) + p(7)$$

$$= {}^9C_5 p^5 q^4 + {}^9C_6 p^6 q^3 + {}^9C_7 p^7 q^2 = \tfrac{4672}{6561}$$

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